RNA-seq recovers a mitochondrial genome that its own assembly does not annotate: introns, expressed ORFs and a 2,135-fold gain in *Pleurotus ostreatus*

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Abstract

Mitochondrial genomes are routinely deposited alongside nuclear assemblies and routinely left unannotated, on the reasonable grounds that they are not what the assembly was for. We show what this costs, and that ordinary RNA-seq is sufficient to repair it. The mitochondrial contig of the *Pleurotus ostreatus* BOM_ss5 assembly carries 13 annotated features across 71,949 bp, of which one is protein-coding. Combining tRNAscan-SE, barrnap, ORF prediction under the empirically confirmed genetic code 4 and homology identification, we recover 53 features: 20 protein-coding genes, 25 tRNAs, 2 rRNAs and 6 introns. Read assignment to the mitochondrion rises 2,135-fold, from 343 to 732,325 reads across 16 libraries, and 40 of 47 quantifiable features are detectably expressed. Split-read evidence identifies six spliced introns totalling 4,332 bp, two supported independently by Rfam covariance models for group I introns; the largest is excised at complete efficiency in all 16 libraries. We further show that raw split-read counts are a treacherous measure of splicing: 50 of 56 recurrent junctions splice at under 5% efficiency, and 46 of those lie inside the high-coverage ribosomal RNA region, where they are alignment artefacts rather than introns. Twenty predicted ORFs have no characterised homologue, of which five are expressed in at least half the libraries and cluster at two loci. The mitochondrion is, in this organism, an appreciable and almost entirely undocumented fraction of what RNA-seq actually measures.

Keywords: mitogenome, *Pleurotus ostreatus*, group I intron, genome annotation, mitotranscriptome, RNA-seq

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All values are generated from the analysis repository and verified by `scripts/32_audit_numbers.py`. Sample provenance and library preparation are described in the companion resource paper. [**AUTHORS**] markers indicate items only the authors can supply.

1 Introduction

A fungal genome submission is judged on its nuclear content. The mitochondrial contig is assembled because the reads are there, deposited because completeness is a virtue, and annotated to whatever depth the pipeline happens to reach — which, for organellar sequence, is often barely at all. This is defensible: the assembly was not undertaken to describe the mitochondrion.

It is nonetheless consequential for anyone who later maps RNA to that assembly. Fungal mitochondria are transcriptionally active, their genomes are intron-rich [1], and in libraries where ribosomal depletion is incomplete they can account for a substantial share of sequenced material. Reads that map to unannotated mitochondrial sequence are not merely uncounted; they are reported in categories — "no feature", "ambiguous" — that most workflows summarise and few interrogate.

Song and colleagues' analysis of the *Bipolaris sorokiniana* mitogenome [1] illustrates what a fully annotated fungal mitochondrion contains: a 137,775 bp genome carrying 12 core protein-coding genes interrupted by 28 introns, 38 tRNAs, and 52 free-standing open reading frames of unknown function. We asked how much of that is recoverable for *Pleurotus ostreatus* from RNA-seq alone, using libraries generated for an entirely different purpose.

2 Results

2.1 The deposited mitogenome is almost unannotated

The mitochondrial contig CM148777.1 of the BOM_ss5 assembly is 71,949 bp with 26.6% GC. Its GenBank annotation carries **13 gene features: 12 tRNAs and one protein-coding gene** (Table 1). A fungal mitogenome typically encodes fourteen respiratory-chain and ATP-synthase proteins together with the small- and large-subunit ribosomal RNAs and a variable tRNA set [2]; the closest well-described agaric, *Agaricus bisporus*, carries fifteen protein-coding genes [3]. Essentially the entire protein-coding and ribosomal complement is therefore missing here.

2.2 A corrected annotation from four independent lines of evidence

We first established the genetic code empirically rather than assuming it. EMBOSS `getorf` yields 51 ORFs of at least 100 codons under translation table 4 against 34 under table 1, confirming that TGA encodes tryptophan — the mould mitochondrial code.

Combining tRNAscan-SE (25 tRNAs, corroborated by aragorn at 22), barrnap for ribosomal RNA, and ORF prediction with DIAMOND identification against Swiss-Prot gives **53 features: 20 protein-coding, 25 tRNA, 2 rRNA and 6 introns** (Fig. 1, Table 1). Identified products include cytochrome b, cytochrome *c* oxidase subunits 2 and 3, NADH dehydrogenase subunits 2, 4 and 5, ATP synthase subunit a, and the small ribosomal subunit protein uS3m. Two products are intron-encoded endonucleases of the aI5 and aI8 families, which is itself evidence of intron content: these enzymes are encoded *within* group I introns.

Every feature in the released GFF carries the evidence that produced it, so a reader who distrusts any one line can discount it without re-deriving the rest.

2.3 Read assignment improves 2,135-fold

Across all 16 libraries, 21,180,726 alignments fall on the mitochondrion. The stock annotation assigns **343** of them. The corrected annotation assigns **732,325**, a 2,135-fold increase, and **40 of 47 quantifiable features are detectably expressed** (Fig. 3a).

The corrected figure is still only 3.5% of mitochondrial reads, and the reason is instructive: barrnap identifies the conserved cores of the ribosomal RNA genes, a few hundred base pairs each, whereas the transcribed ribosomal regions span several kilobases. Annotation completeness is not binary, and the residue here is ribosomal rather than protein-coding.

2.4 Six introns, confirmed by splicing

Split-read junctions were pooled across all 16 libraries and filtered to those seen in at least three libraries with at least ten supporting reads, leaving 56 recurrent junctions of 459 raw.

Six of these splice at greater than 50% efficiency and we regard them as genuine introns (Table 2, Fig. 2). They total 4,332 bp, 6.0% of the mitogenome. The largest, at 6,796–7,232, is supported by 4,782 reads, is present in **all 16 libraries**, and is excised at **complete efficiency in every one**. Two of the six coincide with Rfam covariance-model predictions for group I introns, one at $E = 2.7 \times 10^{-15}$.

Covariance-model search independently identifies nine group I intron candidates on the mitogenome and no significant group II candidate. This matters for interpretation: group II introns encode reverse transcriptases ancestral to telomerase and to non-LTR retrotransposon RTs, whereas group I introns encode homing endonucleases and carry no such connection. The intron-encoded endonucleases recovered above are consistent with an exclusively group I complement, and no inference about retroelement or telomerase biology is available from these data.

2.5 Most recurrent junctions are artefacts, and they are not randomly placed

The remaining 50 junctions splice at under 5% efficiency: for each, the overwhelming majority of reads crossing the donor site do so without a gap. **Forty-six of those 50 lie inside the mitochondrial ribosomal RNA block** (Fig. 2b), where coverage reaches 10^6 and a small per-read misalignment rate produces a large absolute number of spurious gapped alignments.

This has a practical consequence we did not anticipate. Raw split-read counts on the mitochondrion vary 20–34-fold *between replicates of the same tissue*, which invites the interpretation that splicing is condition-dependent. It is not: that variance is carried almost entirely by the artefactual junctions, while the genuine introns splice to completion uniformly. **Splicing efficiency, not split-read count, is the quantity that behaves.** We accordingly make no claim about differential splicing between tissues, and would caution against such claims from comparable data without an efficiency denominator.

2.6 Expressed ORFs without characterised homologues

Twenty of the 51 predicted ORFs have no DIAMOND hit to Swiss-Prot and do not overlap an annotated feature. Six are detected at five or more reads in at least half the libraries; one lies inside a ribosomal RNA block and is discarded because its count reflects ribosomal rather than ORF transcription, leaving **five credible expressed candidates** (Table 3, Fig. 3b). They fall at two loci, near 32.8 kb and 65.2 kb, with overlapping predictions at each, so the number of distinct transcribed units is probably two rather than five.

Absence of a Swiss-Prot hit is weak evidence of novelty — Swiss-Prot covers 41% of this organism's nuclear proteome — so these are candidates for follow-up, not discoveries.

3 Discussion

Three points generalise.

Organellar annotation is a systematic blind spot with a measurable cost. The gap here is not subtle: one annotated protein-coding gene where there are twenty, and a 2,135-fold difference in what can be counted. Any study mapping RNA to a fungal assembly inherits this, and inherits it silently, because unannotated sequence produces no error.

RNA-seq is sufficient to repair it. Nothing here required mitochondrial enrichment, long reads, or a dedicated experiment. The libraries were prepared for a tissue comparison and the mitochondrial signal was, from that perspective, contamination. Where poly(A) selection is incomplete — common enough — the resulting non-polyadenylated material is exactly what makes organellar transcript annotation possible.

Split-read counts are not splicing measurements. The distinction between a junction and an efficiently spliced junction is the difference between 56 apparent introns and six real ones, and between an apparent 20–34-fold biological effect and an artefact of coverage. We would not have separated them without computing the unspliced denominator.

3.1 Limitations

The annotation is computational throughout and no gene model has been verified by RACE, proteomics or targeted sequencing. Protein identifications rest on homology, and the ORF predictions inherit whatever errors are present in the underlying assembly, which is contig-level. Coverage is uneven — median depth 2 outside the ribosomal regions — so absence of expression is weak evidence for any individual feature. The intron catalogue is limited to introns that are spliced in these libraries; a constitutively retained intron is invisible to this approach by construction. Rfam group I models are sensitive but not specific at the thresholds used, and seven of the nine covariance-model hits lack independent split-read support.

4 Methods

Culture, RNA extraction, sequencing and read processing are described in the companion resource paper. In brief, 16 libraries were sequenced as poly(A)-selected 3′-tag libraries and aligned with HISAT2 [4] to *P. ostreatus* BOM_ss5 (`GCA_056149245.1`); alignments were reduced to primary records before analysis.

Annotation. Genetic code determined by comparing EMBOSS `getorf` yield under tables 1 and 4. tRNAs from tRNAscan-SE [5] in organellar mode, cross-checked with aragorn [6]; rRNAs from barrnap [7]; protein-coding genes from `getorf` ORFs identified by DIAMOND [8] blastp against Swiss-Prot [9], retaining the longest ORF where predictions nested. Reconciliation and GFF output by `scripts/40_mito_annotate.py`.

Introns. Junctions extracted from CIGAR `N` operations, pooled across libraries and filtered to ≥3 libraries and ≥10 reads. Splicing efficiency computed per library as spliced reads over spliced plus donor-spanning unspliced reads, requiring at least five informative reads. Group I/II predictions from Infernal [10] `cmsearch` against Rfam [11] models RF00028, RF00029, RF01998, RF02001 and RF02003.

Quantification. featureCounts [12] against the corrected annotation and, for comparison, against the stock GenBank features.

Figures

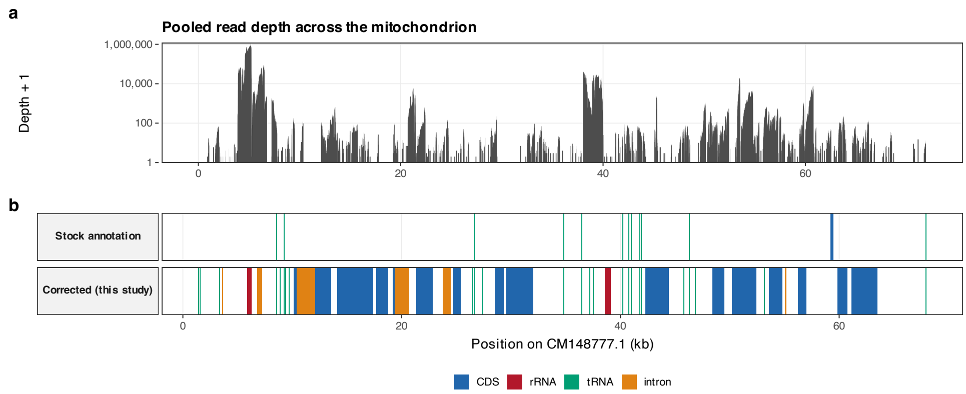

Fig. 1 The deposited mitogenome is almost unannotated. **a**, Pooled read depth across CM148777.1. **b**, Annotated features: the stock GenBank annotation above, the corrected annotation below.

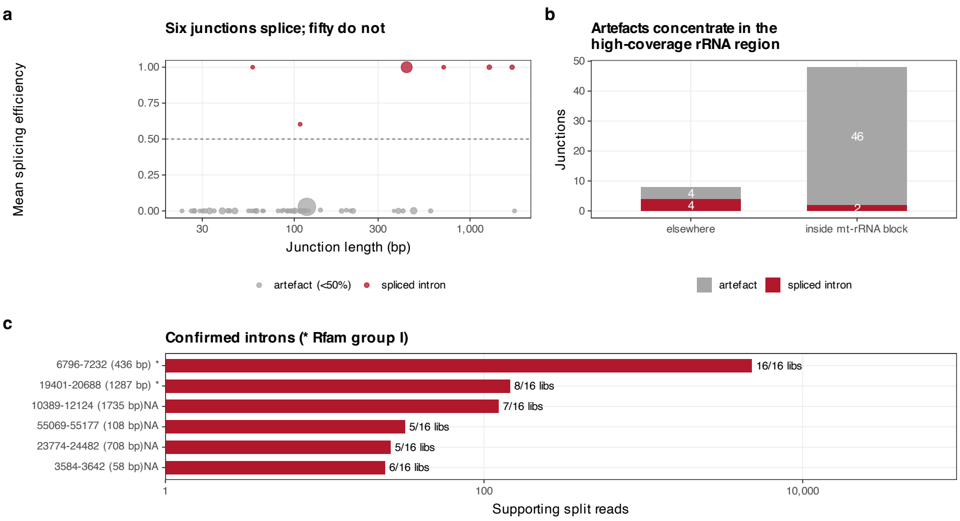

Fig. 2 Six introns among fifty-six recurrent junctions. a, Splicing efficiency against junction length; point size is supporting reads. **b**, Artefactual junctions concentrate inside the ribosomal RNA block. **c**, The six confirmed introns; asterisks mark Rfam group I support.

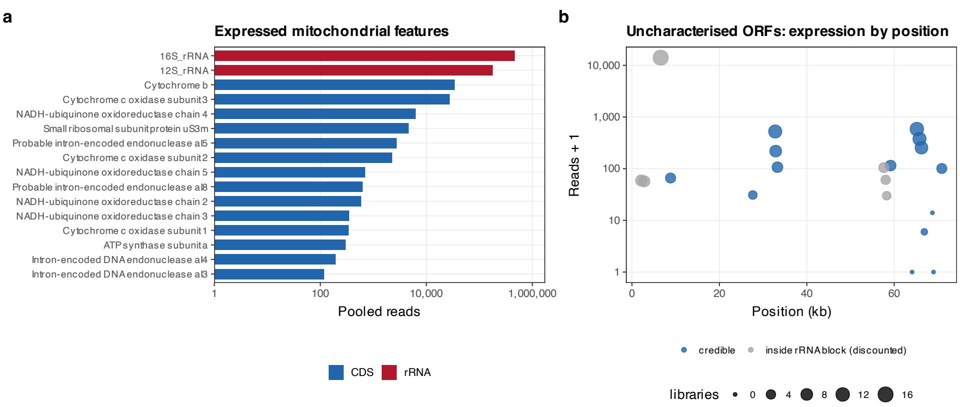

Fig. 3 Expressed mitochondrial features and candidate ORFs. a, Pooled reads for the most highly expressed annotated features. **b**, Uncharacterised ORFs by position; those inside ribosomal RNA blocks are discounted.

Tables

Table 1 Annotation of the *P. ostreatus* BOM_ss5 mitochondrion (CM148777.1, 71,949 bp).

Feature class	Stock GenBank	Corrected (this study)
Protein-coding (CDS)	1	20
Ribosomal RNA	0	2
Transfer RNA	12	25
Introns	0	6
Total	**13**	**53**

Corrected features derive from tRNAscan-SE, barrnap, EMBOSS `getorf` under genetic code 4 with DIAMOND identification, and split-read intron evidence.

Table 2 Mitochondrial introns confirmed by split-read evidence (splicing efficiency >0.5).

Start	End	Length	Reads	Libraries	Efficiency	Rfam model
6796	7232	436	4,782	16/16	1.0	group I (E=2.7e-15)
19401	20688	1287	146	8/16	1.0	group I (E=1.2e-13)
10389	12124	1735	124	7/16	1.0	–
55069	55177	108	32	5/16	0.603	–
23774	24482	708	26	5/16	1.0	–
3584	3642	58	24	6/16	1.0	–

Efficiency is spliced reads divided by spliced plus donor-spanning unspliced reads, averaged over libraries with at least five informative reads.

Table 3 Expressed mitochondrial ORFs with no Swiss-Prot homologue, outside the rRNA blocks.

ORF	Start	End	Length (nt)	Strand	Reads	Libraries
CM148777.1_27	65174	66319	1146	+	584	12/16
CM148777.1_14	32764	33906	1143	+	524	11/16
CM148777.1_33	65782	66177	396	-	376	11/16
CM148777.1_32	66224	66730	507	-	254	10/16
CM148777.1_44	32880	33218	339	-	218	8/16

One further ORF passed the expression filter but lies inside an mt-rRNA block and is excluded: its read count reflects ribosomal RNA rather than ORF transcription.

Data availability. https://github.com/dr-richard-barker/Myco_tissue_RNAseq. The corrected annotation is provided as `refs/BOM_ss5/mitogenome.gff`, with the intron catalogue and ORF tables alongside.

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Competing interests. [AUTHORS: declaration.]

References

[1] Song, N., Geng, Y. & Li, X. The mitochondrial genome of the phytopathogenic fungus bipolaris sorokiniana and the utility of mitochondrial genome to infer phylogeny of dothideomycetes. *Frontiers in Microbiology* **11**, 863 (2020).

[2] Sandor, S., Zhang, Y. & Xu, J. Fungal mitochondrial genomes and genetic polymorphisms. *Applied Microbiology and Biotechnology* **102**, 9433–9448 (2018).

[3] Férandon, C., Xu, J. & Barroso, G. The 135 kbp mitochondrial genome of agaricus bisporus is the largest known eukaryotic reservoir of group i introns and plasmid-related sequences. *Fungal Genetics and Biology* **55**, 85–91 (2013).

[4] Kim, D., Paggi, J. M., Park, C., Bennett, C. & Salzberg, S. L. Graph-based genome alignment and genotyping with hisat2 and hisat-genotype. *Nature Biotechnology* **37**, 907–915 (2019).

[5] Chan, P. P., Lin, B. Y., Mak, A. J. & Lowe, T. M. trnascan-se 2.0: improved detection and functional classification of transfer rna genes. *Nucleic Acids Research* **49**, 9077–9096 (2021).

[6] Laslett, D. & Canback, B. Aragorn, a program to detect trna genes and tmrna genes in nucleotide sequences. *Nucleic Acids Research* **32**, 11–16 (2004).

[7] Seemann, T. barrnap: Basic rapid ribosomal rna predictor. https://github.com/tseemann/barrnap (2013).

[8] Buchfink, B., Xie, C. & Huson, D. H. Fast and sensitive protein alignment using diamond. *Nature Methods* **12**, 59–60 (2015).

[9] The UniProt Consortium. Uniprot: the universal protein knowledgebase in 2023. *Nucleic Acids Research* **51**, D523–D531 (2023).

[10] Nawrocki, E. P. & Eddy, S. R. Infernal 1.1: 100-fold faster rna homology searches. *Bioinformatics* **29**, 2933–2935 (2013).

[11] Kalvari, I., Nawrocki, E. P., Ontiveros-Palacios, N. *et al.* Rfam 14: expanded coverage of metagenomic, viral and microrna families. *Nucleic Acids Research* **49**, D192–D200 (2021).

[12] Liao, Y., Smyth, G. K. & Shi, W. featurecounts: an efficient general purpose program for assigning sequence reads to genomic features. *Bioinformatics* **30**, 923–930 (2014).